

Quantifying Training Load

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To this point in our book, we've described the main physiological targets of high-intensity interval training (HIIT), the parameters that form and fine-tune our HIIT weapons, principles to consider when integrating strength training concurrently in the training program, along with important strategies for maintaining health in the context of HIIT. While our key puzzle pieces are nicely taking shape, one key aspect we haven't yet covered is the process of trying to understand the impact of a given HIIT session on the athlete and the contribution that the HIIT session is making toward solving the overall puzzle.

For anything we might try to understand, like training and performance, we ideally need to quantify its moving parts. Sticking with our military context, the term *surveillance* is used. The main purpose of surveillance for military operations is largely management. As the old adage goes, you can't manage what you can't monitor. Ultimately in training for sport, we often talk about the training load, and that we need to manage it. But before we go too deep into the monitoring of training load, we'd better first try to define it.

Training Load: What Is It?

Rarely have we seen training load clearly defined within the literature. If we were to break the concept of training load down into a first principle compo-

nent, it's difficult. While training load is often discussed in the world of training for sport, it is really more or less an abstract concept that we use to describe theoretically how the physical task involved with training creates stress and fatigue (chapter 7) as well as adaptation and fitness (chapter 3). But here's where it gets tricky. Because it's an estimated abstract concept, where we can never truly reach any precise level of assessment, we can only make approximations of it. That's not to say it's not useful to do, but it's important to acknowledge these limitations from the start. We have to do a bit of calculated guesswork in the area. That means we always need to take our calculations with a grain of salt.

The stress of any theoretical training load can be considered from two different but complementary concepts. Training load can be described from either the external standpoint, based on how much physical work (J) is actually completed in training, or from the internal standpoint, referring to the body's acute physiological response to the external work being completed. The latter aspect, the internal training load (i.e., $\dot{V}O_2$, lactate, RPE) is a display of the body's acute reaction, or stress, to the applied external training load performed (figure 8.1). Whichever way we look at it, this training load stress evokes an input to the physiological system.

In this chapter, we'll set our focus on aspects of the training load itself, while we'll devote chapter 9

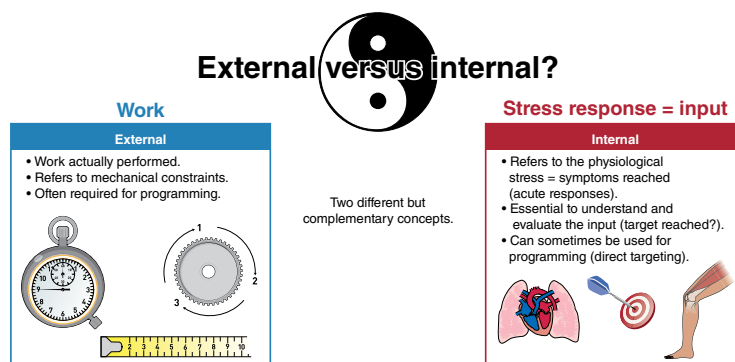


Figure 8.1 Comparing internal (red) versus external (blue) training load concepts. The external training load applied evokes an internal training load stress that provides an input to the athlete's training stress-adaptation system.

to the carryover effects of these training load arsenals, in terms of the damage they inflict or the *response* that the training load stress imparts on the athlete. Beyond these immediate chapters on surveillance, we'll have our expert practitioners show us how they attempt to manage their surveillance systems in the heat of battle. As they'll tell us, this is often easier said than done within the real world of elite sport. Nevertheless, the purpose of these next two chapters is to lay down some groundwork in the area, look to see what might be optimal, consider what is actually practical, and combine science with common sense to give us some surveillance principles we can take to the battlefield of elite sport.

An Aerial View of the Battlefield

If we're just starting out with the process, it might be a good idea to review figure 1.5 to remind ourselves of the importance of context around the sport and individual athlete. What's our individual athlete's characteristics or physiological profile? What is the sport we are talking about? Is it individual or team-based, and what are the associated demands we need to be preparing for?

For the majority of situations, the manipulated variables of an HIIT session outlined in figure 4.5 (i.e., intensity, duration, recovery, etc.) need to be considered first and foremost. Was the completed training in line with what was prescribed? The military would describe such factors as *targeted interceptions*. In military context, it means that these aspects are *marked*—the variables matter—so we need to measure them.

But context is vital. What is the main sport modality we're referring to, and from it can we

easily, noninvasively, reliably, and accurately measure how the parameter evolved over time in an HIIT session, such as the power output or movement speed? How do the internal training load markers evolve during the HIIT session relative to the external training load? From this surveillance, can we gain insight into the likely oxidative, anaerobic glycolytic, and/or neuromuscular responses (i.e., input) to the stress we imparted on our athlete during the HIIT session?

Perhaps one of the most important reasons for surveillance relates simply to the fact that we are all individuals, a factor highlighted throughout nearly every chapter to this point of our book. We all have varying phenotypes, diverse daily psychosocial stresses, different diets, unusual lifestyles, etc. These other stresses, highlighted in chapter 7, can all cause us to respond just a bit differently during the same HIIT workout (i.e., similar external loads but different internal loads). Surveillance and athlete monitoring is an enormous area of research in sports science, with the holy grail thought to be found within the factors that make up the dose-response relationship between training and performance (26).

Once we've collected the data that matters, and we have it in front of us, then what? At first glance, the data collected is more or less just interesting. How do we move from interesting to meaningful and useful, where it becomes a tool that helps us make appropriate decisions toward solving our programming puzzles? For this we require some method of analysis, a method that is as simple and logical to perform as we can make it. We need a system and a process, and it must be relatively easy and practical.

Another important reason for monitoring an athlete's training load relates to the improved probability

of lowering an athlete's injury risk. Given the high cost of injuries, especially in team sports (i.e., ~12.5 million dollars per team in the top-four soccer leagues as just one example (2)) and the strong association between injury incidence and a team's performance in the standings (39), it makes sense to invest time and resources into the initiative. Indeed, the last 15 years have witnessed incredibly rapid development of (micro)technology in the field, which is unlikely to slow any time soon (46).

Figure 8.2 provides a schematic representation of the important factors at play in the training prescription and surveillance process. Beginning with the known target of the session (figure 1.5), we can prescribe training from either the external or internal training load perspective in alignment with our target. Take for example the case of a cyclist who requires a type 3 target for the session. We might prescribe an external training load as 5×3 min at 350 W, 2 min passive recovery, while additionally measuring the associated internal training load attained through heart rate and rating of perceived exertion (RPE). Alternatively, if the cyclist lacks a

power meter, we might aim to hit the same target using an internal training load prescription as 5×3 min intervals at 95% HRmax, 2 min passive recovery. While it is difficult to say that one method of prescription is better than the other, we believe that to efficiently shape the puzzle piece, the more information that we can gather (i.e., both internal and external load measures, and possibly different types of each), the greater the precision.

Thus, in the context of training load programming and our surveillance battlefield, practitioners likely need tools to help them:

Program the INPUT: Define a target reference exercise intensity based on the external load demands (power, speed, movement) or directly target certain levels of internal load ($\dot{V}O_2$, HR, lactate, neuromuscular activity, RPE).

Understand the INPUT: Was the prescribed external training load (i.e., HIIT running as bouts of 4 min at 18 km/h) in line with what was intended in terms of the likely physiological response (internal load response, e.g., 95% HRmax)? This information is required to:

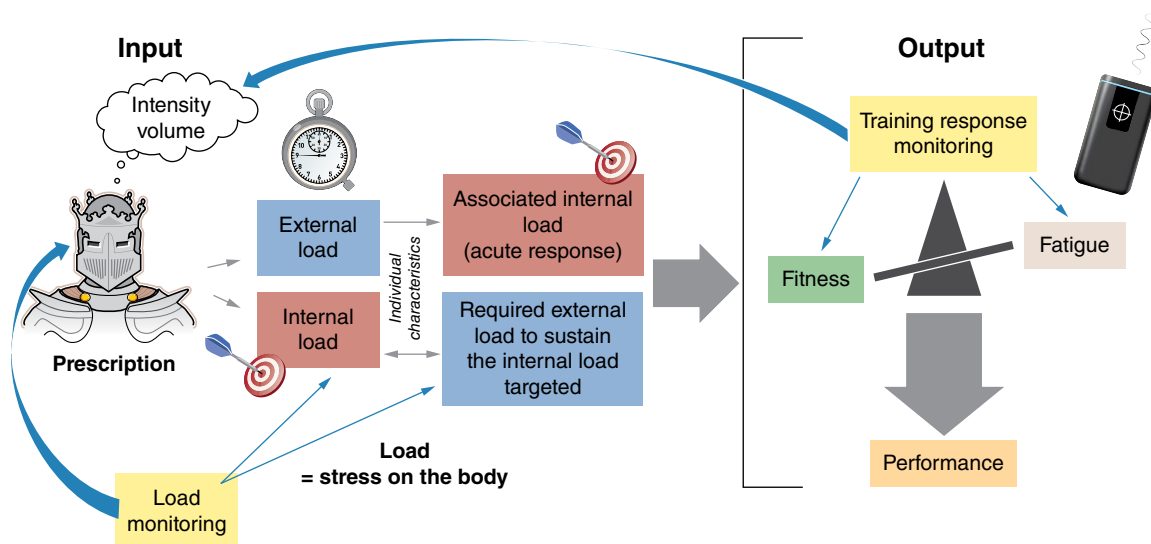


Figure 8.2 An overview of the training load programming and surveillance battlefield. Training, which can be prescribed and monitored as an external training load (blue box) can be seen to impart stress or an input on the athlete, measured additionally as an internal training load (subsequent red box; i.e., HR response to a given power output). Likewise, an internal training load stress can be prescribed and monitored (i.e., 3 min bouts at 95% HRmax) while measuring the related external training load parameters (subsequent blue box, power or speed). The response to the internal training load stress creates the signal for physiological adaptation in the athlete, viewed jointly as both fatigue and resulting fitness. The gain of fitness in the absence of fatigue theoretically enables an athlete to maximize his or her physical performance potential. Blue arrows indicate the retrospective feedback loop that can be used to learn from the surveillance process to adjust subsequent training load prescription.

- Adjust immediate and future external load prescriptions when physiological targets are not reached;
- Adjust immediate and future external load prescription to better fit the puzzle (e.g., cardio-respiratory responses were as expected but anaerobic participation and neuromuscular demands were too high relative to a game scheduled to occur in 48 h); and
- Improve return to full training process with progressive loading in the case of a previous injury.

Understand the OUTPUT: Once integrated into a larger model, the responses or changes to the load can then be used to optimize the process, as will be our focus in chapter 9. This helps us to learn from the past (retrospective analysis showing links with positive versus negative adaptations and outcomes), including physiological adaptations, fitness, and performance; fatigue; and injury.

Targeted Interception Markers for HIIT

Our first priority targeted interception markers within the context of HIIT surveillance need to be those outlined in chapter 4 that describe an HIIT session, and ultimately form the prescribed external training load puzzle piece. As a reminder, these include the:

1. work bout intensity
2. duration of the work bout
3. recovery period intensity
4. recovery period duration
5. number of intervals or series duration
6. number of interval bout series
7. between-series recovery duration
8. between-series recovery intensity
9. total work performed

These all can be measured as either direct or indirect markers of work rate, such as external power output or movement speed, and used to describe the activity in question, to enable us to classify our HIIT weapon (chapter 4) and target type (chapter 5) accordingly. Ideally, a relevant internal training load marker, such as heart rate, used as a proxy measure of $\dot{V}O_2$ (with limitations discussed in chapters 2, 3, and 9), is also captured alongside the external load marker to assess the success of our targeting (or lack thereof).



Figure 8.3 The variety of internal and external sensors currently at our disposal to estimate training intensity, and in turn, training load in athletes.

The following section outlines the primary tools currently used to describe markers of internal and external training load in the context of HIIT. As shown in figure 8.3, there are a number of different ways we can measure both the external and internal training loads, and with (micro)sensor technology on an exponential rise, becoming smaller and smaller, the numbers of integrated and embedded devices are expected to rise substantially over the next decade.

As shown in figure 8.3, there are already a number of sensors today at our disposal. While an important role of the sport scientist is to drive innovation in an attempt to gain a fair advantage over competition, it's vital not to lose sight of important day-to-day operations. As first told to us in 2007 by good friend and road cycling chapter contributor Marc Quod, this push for innovation and integration of the most recent technologies into daily training quantification and monitoring practices should not take us away from effective industry practices, simplicity, validity, reliability, and integrity. In Marc's own words, "Occam's razor always applies." Simply put, no more assessment should be used than is really necessary, as overmonitoring will add distracting noise to your daily smooth operation. Experiment with new emerging technologies with staff and colleagues, but on the battlefield, use only what is needed (29).

Training Load Variables

Training load can be quantified as the product of both the intensity and duration of the exercise bout. What follows first in this section is a description of the different ways we can monitor the intensity of exercise. More details on the load calculations (intensity $\times$ volume) will be described in the last part of the chapter.

As a first port of call, and in the context of a book on HIIT, we need an objective way to be able to describe how an athlete's HIIT session is actually completed. In chapter 2, we outlined the traditional methods that coaches have used to prescribe HIIT, including the RPE-based method, the maximal aerobic speed and power-based approach, the 30-15 intermittent fitness test, anaerobic speed reserve measures, all-out sprint training, the track-and-field approach, the team sport style, and additionally heart rate and power meter–based methods. In the upcoming 20 sport application chapters, you will read examples of how sport practitioners prescribe and monitor HIIT in the context of their sports. As shown in figure 8.2, the chosen method of surveillance will depend on the sport and available technologies, but in any of these cases, we ideally need to get a handle on relevant markers of both internal and external training loads. Take the case of a cyclist for whom we may have a direct measurement of external load via his cycling power output, but we can additionally measure his internal load using heart rate. In the context of sports that are running based (most), we can usually revert to the track-and-field approach using a simple stopwatch, recording the time it takes the athlete to cover a set distance for any given HIIT session to measure external training load, and typically use a heart rate monitor to measure internal load (if relevant). More advanced technological methods now include global positioning–based systems (GPS), semiautomatic cameras, and radio-frequency systems to measure movement speed. As we will discover shortly, in the context of many sports, the movements and associated stresses are complex, and we need to rely on other methods to gain practical insight into an athlete's experienced internal and external training loads.

External Training Load Variables

This section describes the main external training load marker variables of concern in sport and the technology tools currently available to monitor them. Relevant external training load markers ultimately refer

to the mechanical aspect of movement, and the rate at which external work is produced. To be clear, with all of these methods, our aim is to gain insight into the exercise intensity. Variables we can currently measure include an athlete's power output, movement speed, and/or the number or rates of accelerations. The associated external tracking system tools that can be used to monitor these variables include power meters, GPS, local positional systems, semiautomatic video, and linear position transducer technology (4).

Power Output as a Marker of Mechanical Intensity

In the context of HIIT, we really need to get a handle on arguably the most important variable of all, the exercise intensity. Nowadays, power output, or the rate at which work is done, can be measured in watts (J/s), across a number of exercise modes using stationary ergometers, to give us a direct measurement of the exercise intensity. But measurement of power output is no longer exclusive to laboratory-based stationary ergometry. The last two decades have witnessed an explosion in the number of portable measurement devices that can remotely measure, either directly or via estimation, the power produced during exercise completed in the field. Thus, within the context of assessing a target kill for a given weapon (chapter 5), power output or HIIT work rate completed is highly relevant.

Nowhere has the measurement of field-based power output been more successful than in the sport of cycling. In the 1980s, German medical engineer Ulrich Schoberer invented the world's first cycling power meter, consisting of a strain gauge embedded into a bicycle's pedal crank arm. Using the force recorded in the strain gauge, alongside the angular velocity inferred from the cyclist's pedal rate, meant that power output (W) could be accurately measured in the field, just as it was done in the laboratory. Like a GPS can measure speed and total distance across a workout, a cycling power meter provides even better resolution, giving us a direct measure of intensity across any given moment in a ride, in addition to the total work (kJ) applied across any select time period.

Although not without some level of error, Abbiss et al. (1), Quod et al. (55), and others have established the accuracy, reliability, and validity of these devices in the field, and an explosion of related tools and analysis software programs has since followed, capitalizing on improvements in unit size, battery life, and automated wireless data transfer and analysis. Today, systems such as TrainingPeaks, Golden Cheetah, Garmin Connect, and others can automatically receive

and analyze the training performed by a cyclist as soon as the session is completed, and in emerging cases, even in real time. Chapter 15 on road cycling expands on this area and describes how these devices and their software systems are used in the field of professional cycling.

Power output can additionally be measured on stationary ergometers that are specific to an athlete's exercise mode, such as for rowing (62), kayaking (51), and cross-country skiing (48, 49). While power output during these exercise modes can be measured in the field (i.e., via strain gauge and accelerometry), the lack of a constant moment arm length (relative to the cycling crank example) makes accuracy currently challenging, with commercial solutions slow to progress relative to the cycling market. Thus, individual sports outside of cycling today mostly rely on accurate GPS, using reference to world best time or pace (see rowing and cross-country skiing chapters as examples). Additionally, estimations of external power output associated with these alternate forms of locomotion are technically possible via combined GPS and accelerometry systems, but are not without limitations (described below).

Locomotor-Based Rates of Movement With the continued development of the (micro)technology industry, athlete movement tracking is today consid-

ered one of the most important components of load monitoring for both individual and team sports (4). In fact, it is difficult today to find an elite individual sport athlete or team without minimally using one or multiple forms of GPS, semiautomatic cameras, or radio-frequency system tracking during both training and competition (4). While coaches and practitioners may monitor a number of different aspects of training load throughout the week, the monitoring of various aspects of HIIT sequences (see chapters on soccer and Rugby Sevens) is of primary importance for optimizing the overall training impact.

The first aspect to consider when it comes to external tracking systems is that despite the vast array of technologies available today (figure 8.4), at varying costs, the overall reliability (how consistent or repeatable is the measure?) and validity (is it measuring what it's supposed to be measuring?) of most systems tend to be about the same (23). While advances in technology in coming years will undeniably improve these features, our view today is that it may not be the technology per se that matters most but rather the selection of the variables that can seamlessly be collected and how practitioners use them to understand load.

When it comes to selecting the best tracking variables available on the market today, the classification of Gray (21) may be useful. Gray uses three distinct

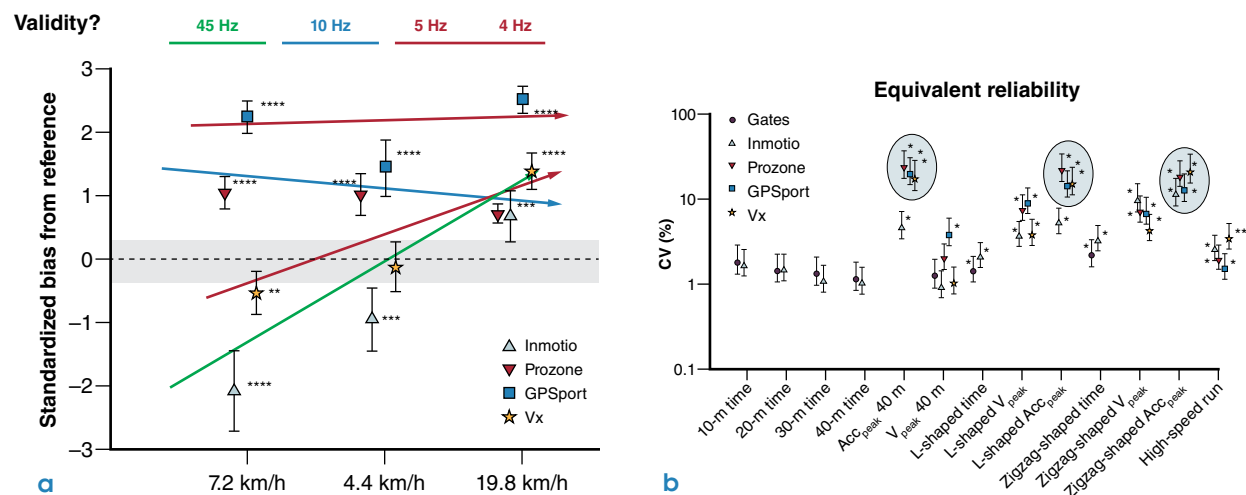


Figure 8.4 (a) Standardized difference in distance covered at various speeds using the different technologies (GPS, semiautomatic video and local positional system). No current technology provides a perfect estimation of actual distance. (b) Reliability of external load tracking variables using current technologies across various drills. The level of reliability is likely similar across the four systems shown. Note the reduced reliability across some of the most important load variables.

Adapted by permission from M. Buchheit, A. Allen, T.K. Poon, M. Modonutti, V. Gregson, and V. Di Salvo, "Integrating Different Tracking Systems in Football: Multiple Camera Semi-Automatic System, Local Position Measurement and GPS Technologies," *Journal of Sports Sciences* 32 no. 20 (2014): 1844-1857, <https://www.tandfonline.com/doi/full/10.1080/02640414.2014.942687>.

levels to describe the capabilities of current tracking technologies (21):

Level 1 reports the typical distances covered in different velocity zones; old-school-type analysis, provided by all technologies; e.g., 345 m of running above 19.8 km/h.

Level 2 reports events related to changes in velocity, including accelerations, decelerations, and changes of directions; provided by most technologies; e.g., 45 accelerations over 3 m/s across a total distance of 233 m.

Level 3 reports events derived from inertial sensors or accelerometers; microtechnology only, so not available with camera-derived systems; e.g., 17 impacts above 6 G, player load of 456 AU, stride variables (ground force load, contact times), stride imbalances (4% reduced impulse force on right leg).

More specifically, to assess HIIT load for run-based work, the key aspects we need to consider are both the intensity of exercise and the overall volume of the session (total distance or TD). Calculating the TD/min may be the best capture of exercise intensity (level 1 type variable). However, specific neuromuscular constraints must also be considered, including high-speed (HS, level 1) stride work involving large hamstring muscle activation, total HS distance in zones (see below), and the total amount of HS work in a given time period (HS/min), as well as time accelerating, decelerating, or changing direction (level 2, summarized as mechanical work, MW, involving thigh muscles (quads and hamstrings), total and MW/min) (20). Level 3 type variables are used more from a training status surveillance perspective, detailed in chapter 9. Finally, metabolic power has also received growing attention (54). This hybrid measure, based on both level 1 and 2 type variables, aims to provide a good estimate of the overall cost of high-intensity actions by combining the actual cost of HS (level 1) and accelerated (level 2) running (54). Unfortunately, however, practitioners are left with a difficult dilemma when it comes to selecting their variables, since the validity and reliability of each is likely to be inversely related to their load monitoring importance, i.e., high-speed running, acceleration/deceleration work, and metabolic power being the least valid and reliable variables (4, 41). Put another way, the variables believed to be best wind up likely being the least useful (18). This does not mean that such variables should not

be monitored, but rather suggests that greater care must be taken when interpreting their differences or changes (i.e., defining a larger, more conservative smallest worthwhile difference or change; see chapter 9) (19).

In addition to the validity and reliability issues outlined, other important limitations of tracking variables must be appreciated as follows (20):

- Acceleration values attained should be directly related to the time window of recording (in general between 0.2 and 0.8 s) and the signal filtering technique used (20, 41). Consensus regarding the time window and filter optimums has yet to be established, however, a simple solution might be to report such values over a single meter of length (i.e., base International System of Units) (38).
- Commercial systems producing this technology often update their data processing techniques (software or unit chipset updates), creating large differences in data output (24). This makes it difficult to maintain historical databases, except in the unlikely case in which a system is not updated.
- The number of available GPS satellites and their random spread (geometric dilution of precision) with respect to time-of-day location variation, or interfering infrastructures (stadium roofs causing partial blockage), result in unclear readings. Unfortunately, few activity reports readily provide this important detail.
- Large discrepancies in GPS distance are known to occur between Doppler and local coordinate techniques. While Doppler tends to be the preferred method, interbrand inconsistencies remain.
- Speed zone acceleration and distance validity are acceleration (5)- and speed (23)- dependent; i.e., validity is lowered with increases in acceleration and speed. Again, variables of most importance are likely the least useful.
- Increased sampling frequency does not always result in better precision or validity (23).
- Large between-unit variations (up to 50%) have been reported, even between units of the same brand (24). To partially counter this limitation, players should always use the same unit, and we have to remain cautious about our interpretations when comparing player data (using larger magnitude thresholds for meaningful differences) (19).

While we were originally excited about the potential of quantifying metabolic power (MP) from GPS readings (22), we have since reconsidered our stance and today question its usefulness in the field as a tool for monitoring elite-level team sport athletes (i.e., Ferraris). This is essentially related to recent research questioning the validity of the MP construct in the context of team sport-specific movements, combined with the fact that the MP calculation represents an incomplete metabolic assumption of internal load due to its broad external load capture.

At least four (16, 22, 42, 63) distinct and independent research groups have now shown locomotor-related MP assessed via either GPS or local positioning system (P_{GPS}) as being largely different from true metabolic demands as assessed using gold standard indirect calorimetry ($\dot{V}O_2$ measures, $P_{\dot{V}O_2}$). P_{GPS} was reported as very largely greater than $P_{\dot{V}O_2}$ during walking (16), yet very largely lower during low-speed shuttle runs (63) and soccer (22)-, rugby (42)-, and team sport (16)-specific circuits.

Despite its great potential, at this point in time the questionable accuracy, validity, and reliability of P_{GPS} prevent us from recommending its use as a surveillance tool for team sport-specific efforts with currently available commercial technology (23).

Considering that the agreement between P_{GPS} and $P_{\dot{V}O_2}$ appears acceptable only during continuous linear running (but not during walking and intermittent changes of direction) (16), the metabolic underestimation consistently reported (16, 22, 42, 63) appears to be arising from the application of a non-specific linear sprint acceleration formula to calculate P_{GPS} (34), which is clearly not suited for team sport-specific running patterns (e.g., rest, irregular step frequency and stride length, turns, upper-body muscle activity, static movements) (22). Additionally, if P_{GPS} were to reflect only locomotor-related metabolic activity (as opposed to a systemic measure such as $P_{\dot{V}O_2}$), what is the real value of knowing this number anyway? Presently, these identified limitations (figure 8.5) highlight the questionable value of monitoring P_{GPS} in team sports. The usefulness of P_{GPS} is also limited with respect to what practitioners should actually be aiming to monitor in the field. Surveillance should include:

1. Overall estimates of internal training load, which in our view is satisfactorily assessed using HR and RPE measures (4); currently, metabolic load measured exclusively using locomotor-related

actions such as P_{GPS} is not comprehensive enough.

2. Precise measures of external load, related directly to specific mechanical muscle group constraints, which have direct implications for training, recovery, and injury risk. However, P_{GPS} is clearly dissociated from actual muscle activation, as shown by the large variations in the P_{GPS} /electromyography (EMG) ratio with running that involves accelerations and decelerations (38). If P_{GPS} is used as a global marker of mechanical work (combining high-speed running and mechanical work), it does not decipher the underlying specifics of the load (figure 8.5). The more logical approach is to use the accelerating, decelerating, and high-speed running distance since these are more representative of the specific muscle group load. As injuries tend to be more related to inappropriate volumes of accelerations (14) or high-speed running (35), there is little evidence to suggest that spikes in overall energy consumption per se play any role in injury etiology.

In conclusion, while there are many ways we can skin the cat when it comes to team sport surveillance, we believe that current best practice for characterizing the external load of a training session is to monitor a session's total distance, high-speed running distance, and mechanical work (and not metabolic power). Table 8.2 (at the end of the chapter) provides individual session examples in two different athletes performing the same run-based HIIT sequence, alongside the internal and external load markers. Further examples can be found in the chapters on soccer, Rugby Sevens, and field hockey.

Linear Position Transducer

Linear position transducer (LPT) technology allows for measurement of the velocity of many types of gym-based movement. These devices typically take the form of a central processing unit attached to the resistance training equipment (such as a barbell) via a retractable, measuring cable (e.g., Gymaware, Tendo) linked to optical readers that measure cable velocity via its linked wheel rotation (40). LPT devices convert physical attributes (the length of the measuring cable or speed of wheel rotation) into electrical signals that record the displacement of the barbell in this case, other resistance training equipment, or even the athletes themselves. LPT devices

P_{met}: underestimates $\dot{V}O_2$ or not?

- Stevens 2014 (slow running shuttle):
 $P_{met} \ll \dot{V}O_2$
- Buchheit 2015 (soccer-specific):
 $P_{met} \ll \dot{V}O_2$
- Twist 2016 (rugby-specific):
 $P_{met} \ll \dot{V}O_2$
- Brown 2016 (team-sport-specific circuits):
 $P_{met} \ll \dot{V}O_2$

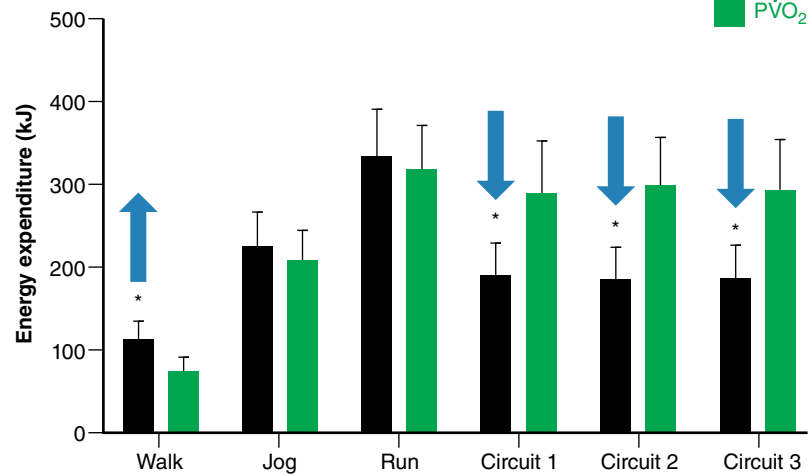
P_{met} versus $\dot{V}O_2$: Task-specific?

Figure 8.5 Four independent studies (16, 22, 42, 63) that have as of today clearly shown the lack of agreement between metabolic power (P_{met}) assessed via either GPS or local positioning system as being largely different from true metabolic demands as assessed using gold standard indirect calorimetry (P_{VO2}). P_{GPS} was reported as very largely greater than P_{VO2} during walking (16), very largely lower during low-speed shuttle runs (63) and soccer (22), rugby (42), and team sport (16)-specific circuits.

Reprinted from <https://martin-buchheit.net/2016/12/01/metabolic-power-powerful-enough-to-drive-ferraris/>.

display live, velocity-based feedback via a display screen or secondary device (such as personal computer or tablet device). The measurement error of such devices has been shown to be very low, and both relative and absolute reliability have been shown to be acceptable. Harris et al. (40) provide a comprehensive review of the technology underpinning this equipment for further reading.

With respect to HIIT programming, encoders can help to monitor the velocity and calculated power output of movements during gym-based circuits (see chapter 6), which can be used as a measure of exercise intensity for type 6 targeting in the same way as cycling power and running speed. There is to date however very little, if anything, published regarding how to best use LPT for prescribing an HIIT session with respect to its neuromuscular component (i.e., type 2, 4, and 5 target responses). This is likely related to the fact that HIIT prescription in the past has generally been based on whole-body movements and/or locomotor modes that involve a large number of muscle groups, and in turn, involve high metabolic demands (e.g., $\dot{V}O_{2max}$ with running or $\dot{V}O_{2max}$ with cycling). Using gym-based movements is less likely (although not impossible) to trigger high systemic metabolic demands. However, the ability to record movement intensity during HIIT is never-

theless important for better understanding HIIT demands, monitoring changes in external load over time, and examining external to internal ratios (see chapter 9).

Internal Training Load Variables

Markers of training load considered internal refer to those that assess how the body has responded acutely during training, as measured by the external training load (figure 8.2). To be clear here, the internal training load markers we will cover in this section are attempts at understanding the intensity of the exercise from an internal measurement. Calculation of the load (intensity $\times$ volume) from our selected internal load marker is covered shortly. Measurement of internal training load is important, as it can offer us solutions to the programming puzzle related to the basic principle of individuality. That is, we all respond differently to any given HIIT session prescription, even one that might be normalized to the individual (i.e., W/kg, % PPO, % $\dot{V}O_{2max}$, etc.). When we measure markers of internal relative to external training loads (expanded on in chapter 9), it gives us direct insight into how the internal machinery of the individual is working to produce his or her external movement speed or power output. Thus,

internal measurements that can be easily monitored become useful adjuncts to record alongside the external training load markers we identified earlier when it comes to putting good surveillance systems around HIIT sessions. While many means of monitoring internal training load (exercise intensity) potentially exist (figure 8.3), the most relevant and practical field-based markers for practitioners are those that align with the key physiologically relevant HIIT parameters we introduced originally in chapter 3. These markers, with known limitations already identified, can be considered proxy readings of the oxidative, glycolytic, and neuromuscular responses during training. In line with these physiological systems, they include heart rate (aerobic demand and cardiovascular work), blood lactate (anaerobic contribution), and markers of neuromuscular function (neuromuscular demands and musculoskeletal strain), respectively. Notwithstanding these direct physiologically aligned variables, we describe further the gold standard holistic marker of internal training load, the session rating of perceived exertions (s-RPE).

Heart Rate

In chapter 2, we described how heart rate is probably the most commonly measured internal training load marker used to measure exercise intensity in the field (17), being well suited to prolonged and sub-maximal exercise bouts. Its challenge of course is the loss of relevance for monitoring the intensity of most HIIT sessions. Indeed, HR alone cannot inform on the intensity of physical work performed above $v\dot{p}\dot{V}O_2\text{max}$ (figure 2.3), representing the large proportion of HIIT prescription (9, 10, 45). Additionally, while HR can reach maximal values (>90%-95% HRmax) during exercise at or below $v\dot{p}\dot{V}O_2\text{max}$, this is not always the case, especially for short intervals (<30 s) (52) and medium-long intervals (i.e., 1-2 min) (61), due to the previously described exercise onset HR-lag phenomenon, with HR lagging well behind the $\dot{V}O_2$ response (figure 2.3) (27). Additionally, HR inertia continues at exercise cessation (i.e., HR recovery), which can lead to overestimation of training load when measured across recovery periods (61).

Notwithstanding its limitations, heart rate is a valuable internal training load marker to monitor in athletes, and while acknowledging that $\dot{V}O_2$ cannot be directly inferred from heart rate, it is the best we have, and at least it tells us about the actual work intensity of the heart. It is simple to take, relatively nonrestrictive for athletes in most sports, with a number of seamless commercial systems now

using automatic uploading via Wi-Fi, Bluetooth, or smartphone direct to software for analysis and communication to external parties.

Use of such a system is shown in figure 8.6, where the simultaneous monitoring of external and internal markers of training load was collected in an elite triathlete during cycling and running short interval sessions (type 4 targets). Note the lag and inertial response of heart rate relative to power and speed, respectively. Nevertheless, the collection of heart rate during such sessions provides good insight into the athlete's health status and function, which will be expanded on further in chapter 9.

Blood Lactate

As we alluded to originally in chapter 3, lactate, or lactic acid when it is dissociated from the H^+ ion, is produced both at rest and during all intensities of exercise from the oxidation of glucose during glycolysis and is an important source of chemical energy (i.e., fuel substrate) during both exercise and in recovery. During steady-state exercise below the lactate threshold, the rate of blood lactate production from glycolysis is matched by the rate of its removal (conversion of lactate to pyruvate via the Cori cycle and lactate shuttle). Lactate produced in working muscle fibers is transported and can flow to adjacent less-recruited muscle fibers to be converted to pyruvate in a process known as the lactate shuttle (15). Here, pyruvate is converted to acetyl-CoA and oxidized through the central energy pathway (figure 3.7). This balanced process is referred to as lactate turnover (11).

Blood levels of lactic acid are increased at exercise intensities above the anaerobic threshold due to the greater ATP demand, and therefore the greater reliance on glycolysis to match the ATP energy requirements. Lactic acid levels begin to accumulate when lactate production rate exceeds its removal rate, and this denotes a high use of the anaerobic glycolytic energy system (type 3, 4, and 5 targets). Changes in pH and bicarbonate accompany these higher lactic acid levels (57). When blood lactate accumulates at a rate in excess of its removal, this causes increases in intracellular acidity, which may cause enzyme disruptions and muscle contraction impairment and ultimately fatigue (11). However, alternative data suggests that higher levels of acidity in the muscle actually assist in the optimization of force production during muscular contraction, making the production of lactic acid during HIIT actually beneficial to muscular contraction (53).

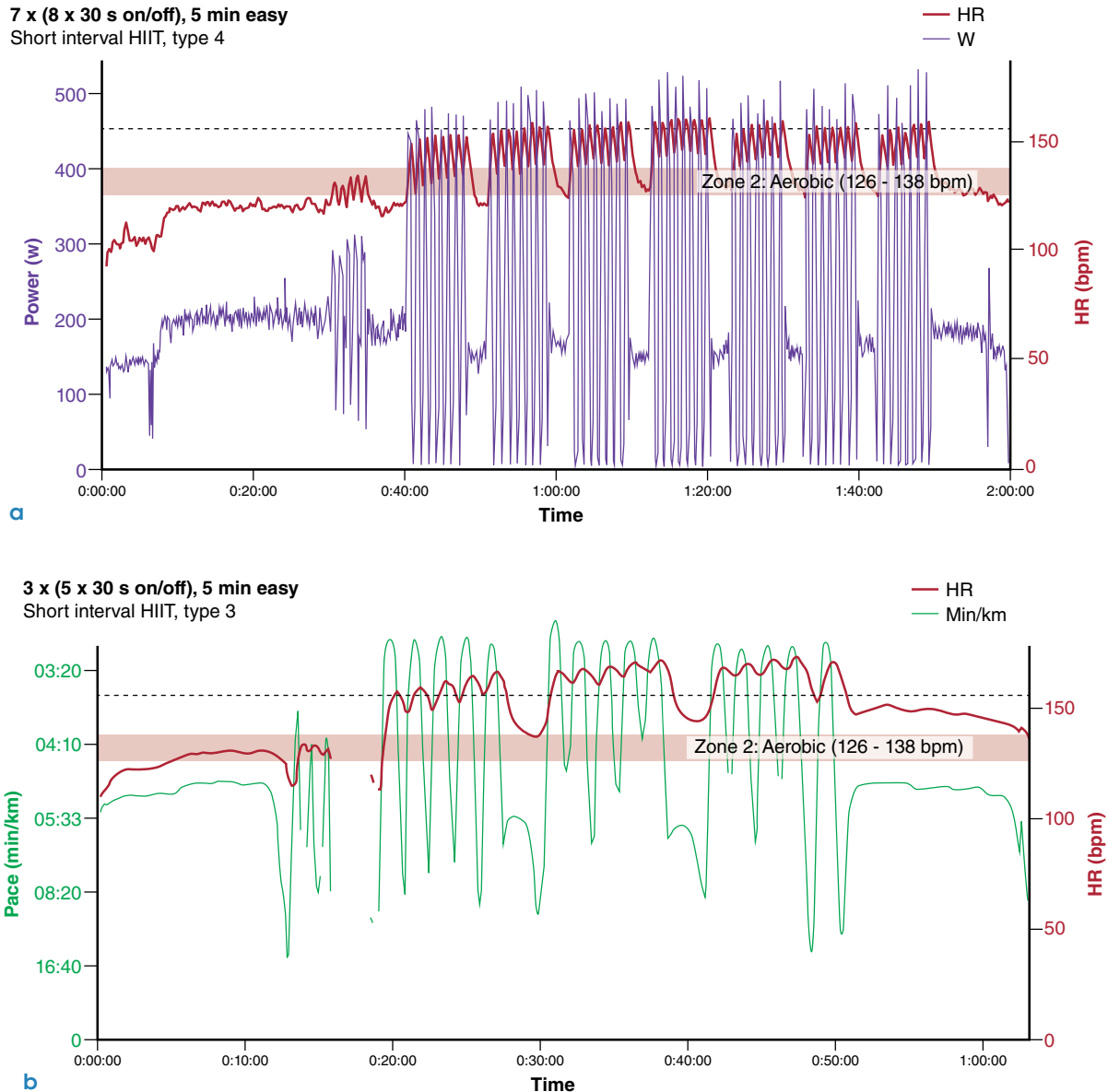


Figure 8.6 Simultaneous monitoring of external and internal markers of training load in an elite triathlete during (a) bike and (b) run short-interval training sessions. Zone 2 (5-zone model) heart rate is highlighted for reference. Note the lag and inertial response of heart rate relative to cycling power (a) and running speed (b).

Regardless of the possible benefits or encumbrances, lactic acid rises during exercise of increasing intensities associated with HIIT, and is a useful variable to monitor when the opportunity is convenient and practical, as it provides insight into the level of anaerobic system participation. In particular, blood lactate is often monitored during progressive exercise testing for endurance athletes during testing phases (see the chapters on rowing, swimming, and cross-country skiing, for example). Coyle et al.

(31) define the lactate threshold as a 1 mmol/L increase in blood lactic acid levels above steady-state levels (figure 3.8). Importantly, however, an individual's lactate threshold will be specific to the exercise intensity (power output or running speed) where the imbalance between the muscle cell's lactate formation and clearance occurs, which depends on training status, diet, and individual characteristics, among other factors. The measurement of lactate can, however, provide a means of

highlighting an athlete's metabolic (aerobic oxidative and glycolytic) responses, and may provide a benchmark for exercise training intensity prescription when incorporated with the associated heart rate, running speed, or power output response. It is important to note that just like the heart rate measurement, the blood lactate measurement will be specific to the mode of exercise due to the neuromuscular blueprint utilized. That is, recruitment of less-trained muscle fibers during unfamiliar exercise modes will alter the lactate response. Thus, the lactate needs to be determined using the mode of exercise specific to the athlete's mode of familiarity.

In the clinical setting, lactate levels are commonly measured using benchtop or portable lactate analyzers. A common benchtop analyzer includes the YSI Lactate Analyzer (12), with the gold standard for lactate analysis thought to be the Radiometer (28). In the field setting, the measurement of blood lactate accumulation (and its rate of accumulation) is the least practical and most annoying internal load measurement for athletes due to its invasive nature. For that reason, in contrast to heart rate, for example, it is currently impossible to measure blood lactate accumulation during every HIIT session. Nevertheless, in some specific circumstances, it can be of great value to confirm or verify the expected anaerobic contribution to a particular HIIT format (see chapter 5), especially when coaches and athletes are seeking to find the best available weapon for hitting their targets within their specific context (e.g., finding no available literature or HIIT data for their given sport or exercise modality). Some of the more common handheld portable lactate analyzers include the Accutrend (36), Lactate Pro (64), and i-STAT (33).

Neuromuscular Function

Assessing neuromuscular engagement (contraction intensity) and musculoskeletal strain during training is now possible via the use of new technology that includes portable EMG systems that measure muscle activity using embedded sensors in clothing, as well as contact time insoles that provide insight into foot work and leg stiffness regulation. While it is difficult to prescribe actual HIIT sessions using this new information, the monitoring of such responses is helpful for understanding whether actual physiological targets have been attained (i.e., type 2, 4, and 5 targets). When such technology is not available, indirect ways we can gain insight into the neuromuscular load of a session is twofold. The first approach is to use typical external load measures

as a proxy of the actual neuromuscular load, since it can be assumed that the greater the high-speed running, mechanical work, or mean cycling or rowing power of the HIIT session, the greater the neuromuscular engagement (i.e., figure 8.6). The second option is to examine pre- versus post-session changes in neuromuscular-oriented performances and/or adjustments during standardized tasks (refer to neuromuscular sections, chapters 3, 4, and 5). While possible post-activation potentiation mechanisms can sometimes occur, it is logical to think that the greater the impairment of neuromuscular performance post-HIIT, the greater the neuromuscular load attained in the session.

Session Rating of Perceived Exertion

The session-RPE method for quantifying exercise intensity was developed originally by Foster et al. (37) using a modification of the Borg scale (13). Instead of asking athletes how they are feeling at any specific point during exercise, about 30 min after they finish their workout athletes are asked the simple question, "How hard was your workout?" and typically use a 1 to 10 scale. This session RPE (sRPE) can then be interpreted as a global proxy of exercise intensity during the session. The reliability of sRPE is only moderate, but its practicality largely outweighs this limitation (43, 58). It is also worth noting that from our experience, accuracy and reliability improve with an athlete's use of the method over time.

Summary of Targeted Interception Markers for HIIT

In summary, markers of internal training load are those that can be used to assess how the various systems of the body respond acutely during training. While we have referred to markers of internal training load, we need to remind ourselves that in actuality we have been referring to markers used to assess the degree of exercise intensity, with direct calculations of load to follow. In line with our three key physiological targets discussed throughout the book, we can gain understanding of the likely internal oxidative load response through measurement of heart rate (notwithstanding limitations), insight into the anaerobic glycolytic response through measurement of blood lactate, and in emerging cases, various means of understanding the neuromuscular and musculoskeletal involvement that were described. Table 8.1 shows that the main internal training load markers of intensity, including heart rate, lactate, and sRPE,

Table 8.1 Typical Five-Zone Training Intensity Model, Including Percentage of Heart Rate Maximum, Blood Lactate Concentration, Rating of Perceived Exertion (RPE), and the Typical Effective Work Time in Zone for Endurance Athletes Performing a Polarized Training Model

These various markers of intensity are more or less correlated, with some exceptions.

Intensity zone	Heart rate (% max)	Lactate (mmol/L)	RPE	Typical effective work time within zone
1	60-72	0.8-1.5	1-2	1-6 h
2	72-82	1.5-2.5	2-3	1-3 h
3	82-87	2.5-4.0	3-4	50-90 min
4	88-93	4.0-6.0	4-6	30-60 min
5	94-100	6.0-10.0	7≥8	15-30 min

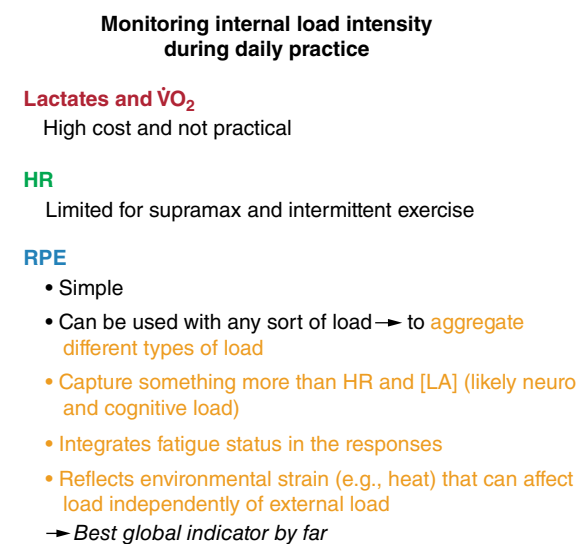


Figure 8.7 Pros and cons of the various intensity markers practitioners can monitor to calculate internal training load.

are more or less correlated. While all markers of internal load surveillance have their place, depending on sport context, the global marker of internal training load (intensity) described was that of the session RPE, which can inform on multiple measures of global internal training load. Certain markers of intensity are better choices than others to measure, with specific reference to the practical element, and a summary of the pros and cons of each are shown in figure 8.7.

From Exercise Intensity to Exercise Load

Irrespective of the load marker considered, we acknowledge that quantification of training intensity

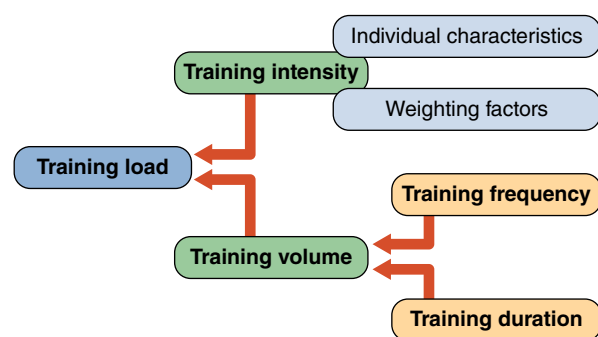


Figure 8.8 Training load is the product of the training intensity and the training volume.

in terms of both external and internal load can be confusing, but let's move on. To this point in the chapter, we have outlined the primary means of assessing markers of exercise intensity from both external and internal standpoints. It's now time for us to make some load calculations. To do this, we need to appreciate that exercise stress (load) is not solely due to the exercise intensity but will additionally relate to the duration of the exercise intensity. The training amount (including the duration and frequency) of an exercise is also referred to as the training volume. Combined, training load can generally be considered the product of the training intensity and the training volume (figure 8.8).

Speed/Power-Based Intensity Zones and External Load

To compute an external training load measure, practitioners will generally combine markers of intensity (i.e., speed or power data) with the actual time spent exercising or distance covered across various (3 to 7) preestablished intensity zones. While it is difficult

to find a sound rationale for all available training zones, practitioners tend to use both absolute (arbitrary) and relative (individual) key intensity benchmarks that correspond with meaningful training intensity bandwidths of relevance, in accordance with the specificities of the sport. For example, a five-zone power output model of training might be used in an Ironman distance triathlete (see triathlon chapter) as the model can generally capture the time in relevant training zones appropriately. However, this model is unlikely to have the resolution of interest for a cyclist with a sprinting role or focus (see cycling chapter), where the model used might be more likely to involve seven zones. While it makes sense to believe that the use of individual zones would better capture an understanding of exercise intensity at the individual level, it requires testing the athlete, which is not always feasible. The more common procedure used by most coaches across sport is the arbitrary method. The following speed zones are used in both the literature and in the field:

ABSOLUTE AND ARBITRARY SPEED (MAINLY TRACKING DATA IN TEAM SPORTS)

- 14.4 km/h
- 18 km/h
- 19.8 km/h
- 25.2 km/h

INDIVIDUAL (BOTH INDIVIDUAL AND TEAM SPORT ATHLETES)

- Maximal power output (55) or sprinting speed (50)
- Maximal aerobic power (55) and speed (50)
- V_{IFT} (59)
- Speed reached at the end of the Yo-Yo IR1 test (25)
- Speed/power associated with LT_2/VT_2 (3, 32)
- 80% of maximal aerobic power or speed (50)
- Speed/power associated with LT_1/VT_1 (60)

Internal Load-Related Intensity Zones, the Training Impulse, and Internal Load

For the range of endurance-related sports, most national sport governing organizations typically use a guiding intensity scale based on ranges of maximum heart rate and blood lactate concentrations at

associated work rates for given durations. Typically, aerobic endurance training in the intensity range of ~50% to 100% of $\dot{V}O_{2max}$ is divided into five intensity zones shown in table 8.1 (60). Unfortunately, this traditional approach, using a standardized intensity scale, fails to account for individual variation in the relationship between heart rate and blood lactate concentration; activity-specific variation, such as the tendency for maximal steady-state concentrations of blood lactate to be higher in activities that engage less muscle mass (7, 8); or level of neuromuscular engagement as previously mentioned. However, for the practitioner, these potential sources of error seem to be outweighed by the improved communication that a common scale facilitates between coach and athlete, as well as across sport disciplines. Speaking the same training intensity language may be particularly important for improving the link between training intensity prescription from a coach's and an athlete's perspective (60).

In chapter 3, we defined three physiologically based demarcation points based on ventilatory or lactate thresholds (figure 3.8). While not directly comparable with the five-zone model, what is typically identified as lactate threshold intensity, or 2 to 4 mM blood lactate concentration, corresponds well in practice with the intensity zone demarcated by the first and second ventilatory turn points (VT_1 , VT_2). So, for practical purposes, the three-zone model and five-zone model (numbered 1-5, table 8.1) are superimposable, with the lowest and highest intensity zones in the three-zone model further divided into two zones. Which intensity zone approach is best? For the young or inexperienced endurance athlete, using a three-training intensity zone model has the advantage of simplicity and makes communicating training intensity goals more straightforward. For the experienced athlete, the five-zone model provides a more precise description of training intensity distribution that may be important for communicating subtle changes in training prescription (60).

Figure 8.8 shows simply that training load can be considered as the product of training intensity and training volume. However, those of us familiar with HIIT know all too well that when intensity reaches the highest levels, the fatigue it generates is considerable. It doesn't take very long for an HIIT session to make a large impact on training load. Thus, the problem with the simple calculation (figure 8.8) is that it does not differentiate between intensity levels of training. Banister (6) was the first to recognize this and noted that blood lactate levels rise

exponentially during intense exercise. Using the fractional elevation in heart rate or heart rate recovery plotted against the blood lactate, Banister (6) was able to gauge a weighting factor (y) to reward the hard work.

Training impulse (TRIMP) was the term Banister used to refer to the integration between time, intensity, and his relative weighting (y) of the exercise intensity to describe the exercise dose into a single number as follows:

$$\text{TRIMP} = \text{time} \times \Delta\text{HR} \times y$$

The TRIMP has been used in many research studies as a means of quantifying training load, and it is used extensively throughout endurance and team sports. For example, when the mean heart rate accrues time in a higher heart rate zone, the TRIMP produces a larger number due to the higher weighting factor of the exercise intensity. While these weighting factors attained from the study have been generalized to the whole population, it must be acknowledged that they are really only reflective of the sample population used in the study, given the blood lactate response is individual. As well, average heart

rate does not represent the large fluctuation of activity seen in intermittent activities such as team sports.

During low- to moderate-intensity exercise, the heart rate and blood lactate response is linear. However, when intensity increases, blood lactate rises exponentially. Thus, for a more accurate TRIMP, we can use knowledge of the individual blood lactate and heart rate response to calculate an individualized TRIMP (iTRIMP) for every heart beat during an exercise session before summing up all values. Figure 8.9 shows the relationship between iTRIMP and performance in distance runners (47).

Anaerobic Glycolytic and Neuromuscular Load

As just described, and in the lactate section earlier (internal training load variables), blood lactate has a number of moving parts to it (Cori cycle, lactate shuttle, etc.), as well as rising exponentially with intensities above the lactate threshold, making assessment of an anaerobic glycolytic load variable difficult, especially from a practical standpoint. While we are aware of companies developing continuous

Exercise average HR (%HR reserve) $\times$ duration

$$\text{Load} = 78\% \times 4 \text{ min} = 312 \text{ units}$$

Time in zones

(Sum of all time in zone (min) $\times$ zone coefficient)

Edwards (panel A)

- 50–60% HR max = coef: 1
- 60–70% = 2
- 70–80% = 3
- 80–90% = 4
- 90–100% = 5

$$\text{Load} = \text{sum} = 11.8 \text{ units}$$

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- < VT1 = coef: 1
- VT1 < HR < VT2 = 2
- > VT2 = 3

$$\text{Load} = \text{sum} = 9 \text{ units}$$

TRIMP (panel B)

Average HR (%HR reserve) $\times$ duration (min) $\times$ weighting factor

$$\text{Load} = 78\% \times 4 \times 0.24^{(3.4 \times 0.8)} = 10.6 \text{ units}$$

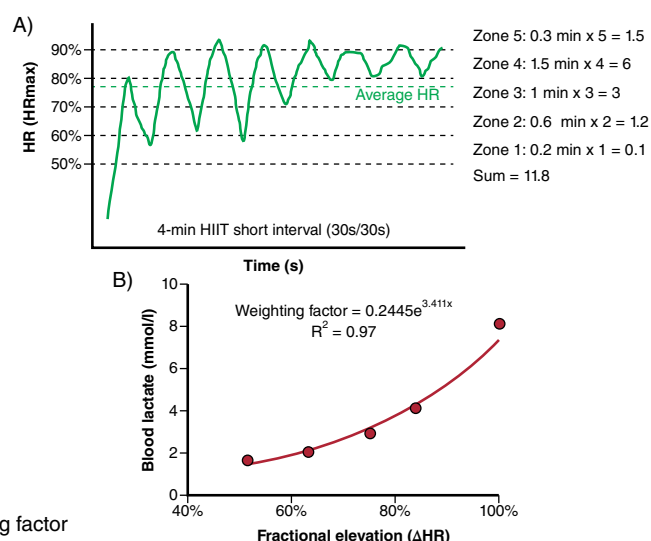


Figure 8.9 Example of the different approaches to measuring cardiovascular load based on heart rate responses. The training impulse approach, originally proposed by Banister (6) was later adapted by Manzi et al. (47), using the individual relationship between blood lactate and heart rate responses to incremental exercise to define the weighting factor.

Graph B: Reprinted by permission from V. Manzi, C. Castagna, E. Padua, M. Lombardo, S. D'Ottavio, M. Massaro, M. Volterrani, and F. Iellamo, "Dose-Response Relationship of Autonomic Nervous System Responses to Individualized Training Impulse in Marathon Runners," *American Journal of Physiology - Heart and Circulatory Physiology* 296 no. 6 (2009): H1733-H1740.

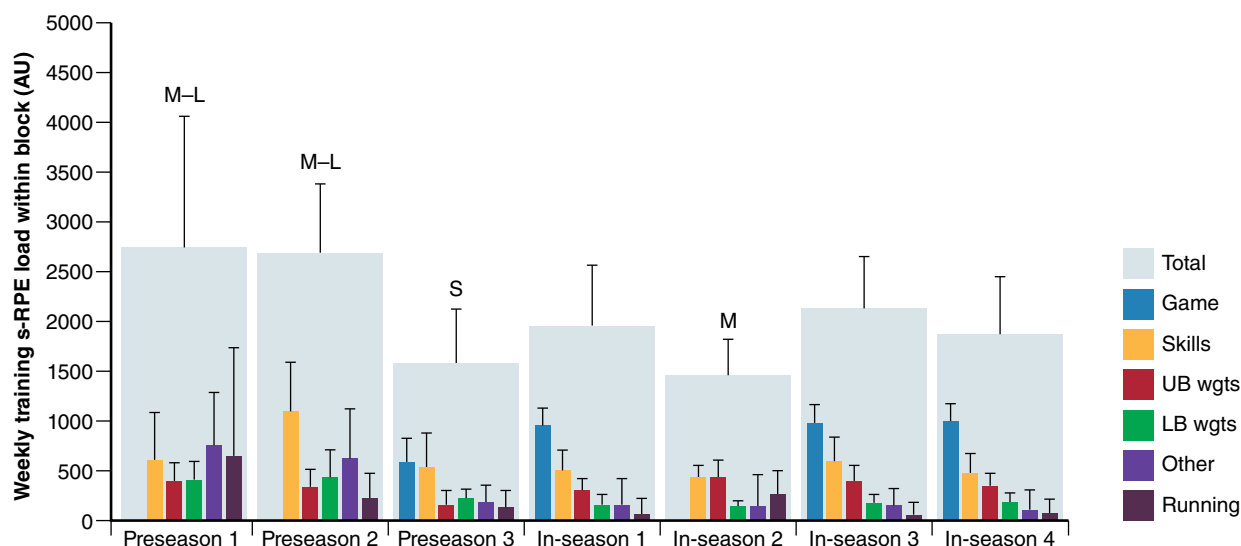


Figure 8.10 Training distribution expressed by RPE per week within block for weekly total load (large bar) and all modes (small bars), mean $\pm$ SD. UB: upper-body; LB: lower-body; AU: arbitrary units; M: moderate standardized difference versus in-season 1, 3, and 4; L: large standardized difference versus preseason 3 and in-season 2; S: small standardized difference versus in-season 1, 3, and 4.

Adapted by permission from D. Ritchie, W.G. Hopkins, M. Buchheit, J. Cordy, and J.D. Bartlett, “Quantification of Training Load During Return to Play After Upper- and Lower-Body Injury in Australian Rules Football,” *International Journal of Sports Physiology and Performance* 12 no. 5 (2017): 634-641.

blood lactate monitors that work using a subdural microsensor, until such technology is realized, calculation of an internal anaerobic glycolytic load variable (intensity $\times$ volume) remains out of the picture. Likewise, with neuromuscular load, while the technologies described earlier in the related section provide promise for the future, as of today we are not able to be confident using their inputs to calculate a neuromuscular-related load variable. Until more progress in the area occurs, the main means of assessing anaerobic/neuromuscular contribution to HIIT remains in blood lactate concentration and accumulation (e.g., 9 mmol/L) and rate of accumulation (e.g., 5 mmol·L⁻¹·5min⁻¹; see chapter 5).

RPE Load

As described previously, the session rating of perceived exertion (sRPE) developed by Foster et al. (37) is among the most commonly used measures for quantifying internal workloads in elite sports. This simple approach involves multiplying the athlete’s RPE for a given session by the duration (volume) of the session (in min), to derive a training load in arbitrary units. As such, a session rated as hard (sRPE = 5) over 90 min would be attributed with a

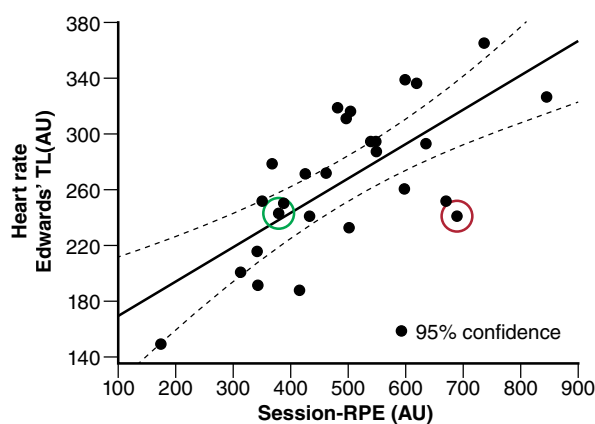


Figure 8.11 Correlation between mean team session-RPE-based training load and HR-based training load (Edwards TL) from 27 training sessions in young soccer players ($r=0.71$, $P<0.001$). Note that while there is a large correlation in overall, there are some large individual variations too. For example, a session with an Edwards load of 230 A.U. can be associated with sRPE values from 400 to 700 A.U. (likely reflecting different levels of neuromuscular load for a similar cardiovascular load).

Reprinted by permission from F.M. Impellizzeri, E. Rampinini, A.J. Coutts, A. Sassi, and S.M. Marcora, “Use of RPE-Based Training Load in Soccer,” *Medicine & Science in Sports & Exercise* 36 no. 6 (2004): 1042-1047.

KEY POINTS: LOAD

These dissociations highlight the fact that each load marker may be specific to a biological system, which makes load monitoring easier and more precise in practice. As conducted by most practitioners in the field (see sport application chapters), by using a combination of simple field-based measurements, such as HR, sRPE, and locomotor data (power meters and GPS), we can characterize the related acute cardiovascular load (HR), neuromuscular and musculoskeletal load (indirectly from power meters, accelerometers, or GPS readings), and overall training load (sRPE). Doing so allows us to see which system is receiving the greatest strain, and to better understand and program the different types of stress, which is needed for training different systems simultaneously for enhancing adaptations and limiting interferences (see chapter 6 and the chapter on soccer). Table 8.2 shows how the load of an HIIT session can be estimated using both internal (HR, lactate, and RPE) and external (locomotor demands via GPS) measures of training load.

Table 8.2 Load Reports Showing External and Internal Load Using Relative and Absolute Thresholds for Two Athletes Performing the Same Run-Based HIIT Sequence

Both athletes performed the same run-based HIIT sequence (8 min, 15 s straight-line running at 20 km/h (83 m)/15 s passive recovery + 2 min passive + 6 min same sequence), HIIT type 4. Note that in absolute terms both athletes performed the same external load (which in fact was prescribed). However, the internal load responses (average HR, rate of blood lactate accumulation, sRPE, TRIMP, Edwards HR load, RPE load) tend to be greater for athlete B. Note that the greater internal load could be somewhat predicted when looking at external load based on a relative speed threshold (i.e., $D > V_{IFT}$). Note also the small difference in estimated cardiovascular load between TRIMP and Edwards methods, likely related to differences in individual HR profiles.

High-speed running $D > 19.8$ km/h, m)	$D > vV_{IFT}$	Mechanical Work (A.U)	Heart rate (% HRmax)	HR-TRIMP	Edwards HR load	Rate of lactate accumulation (mmol.L.5 min)	sRPE	RPE IOAD
1257	1257	6	90	19.5	52	4	7	112
1262	1698	8	91	20.1	48	4.5	8	128

V_{IFT} : maximal speed reached at the end of the 30-15 intermittent fitness test; see chapter 2. $D > V_{IFT}$: distance covered above V_{IFT} .
 V_{IFT} : 20 km/h for athlete A, 19 km/h for athlete B. Mechanical work: overall measure of acceleration, deceleration, and change of direction work (18).

Adapted from M. Buchheit, "Player Tracking Technology: What If We Were All Wrong?" presented at 2nd Aspire Sport Science Conference on "Monitoring Athlete Training Loads: The Hows and Whys," Doha, Qatar (February 2016).

load of $5 \times 90 = 450$ units, while a recovery session rated as easy ($\text{sRPE} = 2$) over the same duration would receive 180 units. One benefit of this approach is that it can be used to aggregate and quantify the various training modalities undertaken by team sport athletes and other mixed-training groups (see the chapter on combat sports, for example). An example of how different training loads arising from resistance training or other sport-specific conditioning and skill sessions can be compiled together with the sRPE method is shown in figure 8.10 from Ritchie et al. (56). In addition, the sRPE method has been shown to relate favorably with objective load measures including heart rate, blood lactate (30), and other factors, such as collisions. Thus, the sRPE method represents an inexpensive and highly practical tool for the monitoring of training loads in diverse sport settings (see combat sport chapter, figure 11.7, for example).

Assessing Training Load in Practice

While there is typically the tendency for all load types (internal versus external) or methods (absolute versus relative thresholds) to be correlated, those correlations are never perfect. For example, there is a tendency for RPE to reflect cardiovascular strain (13), but on an individual session basis, there is often dissociation between RPE and heart rate responses (i.e., when the session involves substantial neuromuscular loading coupled with low cardiovascular strain; e.g., type 5 target). This indicates that the RPE measure is telling us something more than just heart rate. For example, Impellizzeri et al. (44) applied the session-RPE method proposed by Foster et al. (37) in young soccer players across 27 different training sessions to compare internal training load calculations. Relationships with heart-rate-based training load calculations were assessed. As shown in figure 8.11, the heart-rate-based Edwards training load calculation was well correlated with the session-

RPE method. However, there were distinct outliers across the comparison, reflective likely of these individual dissociations mentioned, reflecting the fact that different stressors outside of cardiovascular load were impacting the athlete (i.e., neuromuscular, cognitive-mental-emotional, thermal; figure 8.7). Thus, heart rate and sRPE markers collectively provide complementary surveillance as they together give insight into the effect a training session likely had on more than just a single physiological target.

Summary

In summary, we have shown the importance of putting surveillance systems in place across HIIT sessions through monitoring the stress of training. The stress of training is theoretically calculated as the training load, which can be assessed using markers of external (outside the person) or internal (inside the person) exercise intensity, which is subsequently used to estimate exercise load when integrating exercise volume. Our main methods of monitoring external training load included measures of power output and locomotor activity (power meter and GPS), while internal monitoring loads can be assessed using markers of oxidative metabolic and cardiovascular strain (through HR), blood lactate, and various indices of neuromuscular and musculoskeletal strain. The global marker of exercise training load, and the one that takes advantage of the athlete's own inbuilt sensor of stress, the brain, was termed the sRPE. In the name of application and practicality, and to apply Occam's razor (29), the simple variables in the field are usually enough, with our top three including power or GPS, heart rate, and sRPE. The monitoring of all three provides comprehensive surveillance over the ACTUAL demands, which in addition to providing important programming clues, permits the important assessment of internal-to-external load ratios to be used in the context of training status surveillance, discussed next in chapter 9.